Open Source Licenses: Definition, Types, and Comparison

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SHARE

Have you ever used software that you can change and share with others for free? This is made possible by something called an "open-source license." But, have you ever wondered how this and why it's important?

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Open-source licenses let developers share their source code with the world while still protecting their intellectual property and controlling how it's used. It also helps everyone understand the rules for using and changing the software. These licenses are a big part of the open-source world and

So, what exactly is an open-source license and how does it impact you? What types are there and what potential open-source license risks can you face?

What is an Open-Source License?

Open-source software license definition says that these are legal agreements that determine the rules under which users can access and modify the code of open-source software. These licenses typically allow users to use, change, and distribute the software freely, as long as they follow certain conditions outlined in the license.

Open-source software licenses provide a legal framework that protects both the rights of the software creators and the users. In simple terms, they are like rules for using and sharing free software.

Open-Source Definition

To make sure that open-source software licenses are clear, let's go one step back to briefly describe **open-source meaning in general**.

The main distinction can be made between open-source and proprietary software. With proprietary software, only the company that owns it has access to the source code and can see and modify it. It is protected by a proprietary software license. On the contrary, open-source code is publicly accessible and distributed along with the software. It is made available to everyone and can be modified and improved by anyone with the necessary skills. The idea behind open source is to promote collaboration and innovation in software development.

> **Proprietary software** is owned by an individual or a company. Different types of software licenses restrict it from being freely used, distributed, or modified by others. The proprietary code is typically not available to the public, and users are usually required to agree to a license agreement that sets limits on the use of the software. This type of software is typically sold commercially and can only be used under specific terms set by the owner.

What's the Difference Between Open-Source and Public Dom

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Before diving any deeper into open-source licenses, let's make one more distinction clear - the one between open-source and public domain. These are two concepts related to the distribution and usage of software and other works.

Public domain includes works that are not restricted by copyright and can be used, modified, and distributed without restriction. This can happen when a copyright has expired or when the creator chooses to give up their rights to the work.

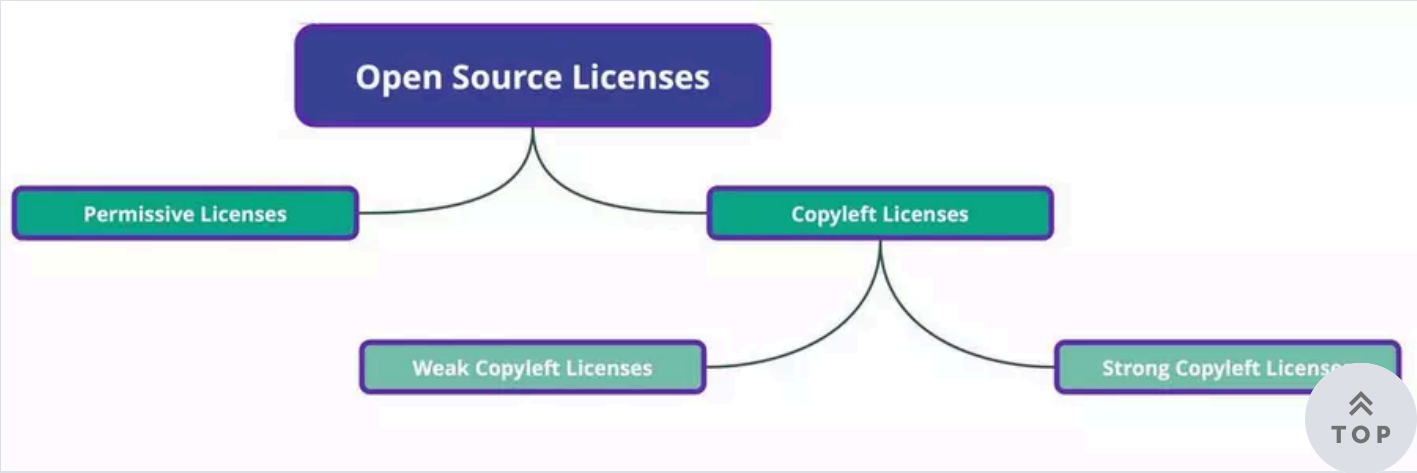
The main difference between the two is that open-source has a license that sets rules for usage, while public domain has no restrictions at all. Both concepts promote sharing and collaboration, but open source provides a more structured framework for it.

The main difference between *open source* and *public domain*

	Open Source	Public Domain
Is there a license attached to software?	✅	❌

Two Main Open-Source License Categories: Copyleft and Permissive

Based on the obligations and limitations imposed on users by the license, there are two main open-source license categories - **copyleft** and **permissive licenses**.



Open-source licenses categories

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Actually, copyleft does exist, and as you can guess it is the opposite of copyright. While copyright restricts the use and distribution of a work, copyleft licenses allow that as long as the terms stated are obeyed.

Imagine you wrote a cookbook and you have the copyright to the book. This means that only you have the exclusive right to use, sell, and distribute the book. No one else can sell or distribute copies of your book without your permission.

Now, imagine that you wrote an open-source cookbook and released it under a copyleft license. This means that anyone can use, modify, and distribute the book for free, as long as they also share their modifications and any derivative works under the same copyleft license. In this case, the book becomes a collaborative effort, with everyone adding and improving the recipes. The goal of copyleft is to ensure that the book remains freely available and that future modifications are also freely available.

As explained in this example, copyright gives the creator control over their work, while copyleft encourages collaboration and sharing. Let's have a closer look at each of them.

Copyleft Definition

Open-source software licenses in this category allow for the free distribution and modification of a work, as long as any derivative works are also licensed under the identical license. It is meant to ensure that the work remains open and freely available for future use, modification, and distribution. To put it another way, your modifications must remain just as open and accessible to everyone with identical rights and privileges afforded by the initial version.

These open-source software licenses can be further divided into:

1. Strong Copyleft Licenses

If a source code is protected by a strong copyleft license, then the derivative software needs to be publicly available under that license as well. This includes all linked libraries and components within the software. For example, if someone modifies and distributes software under a GPL license, they must also make the source code available and license their modifications under the same GPL license. So if you plan to develop software and offer it as proprietary, these are open-source licenses to avoid.

2. Weak Copyleft Licenses

The requirements of a weak copyleft license are similar to those of a strong one, but they apply to a limited set of code. This open-source license only requires that the source code of the original modified work be made publicly available, while the rest of the code used together with the work doesn't have to be published under the same license.

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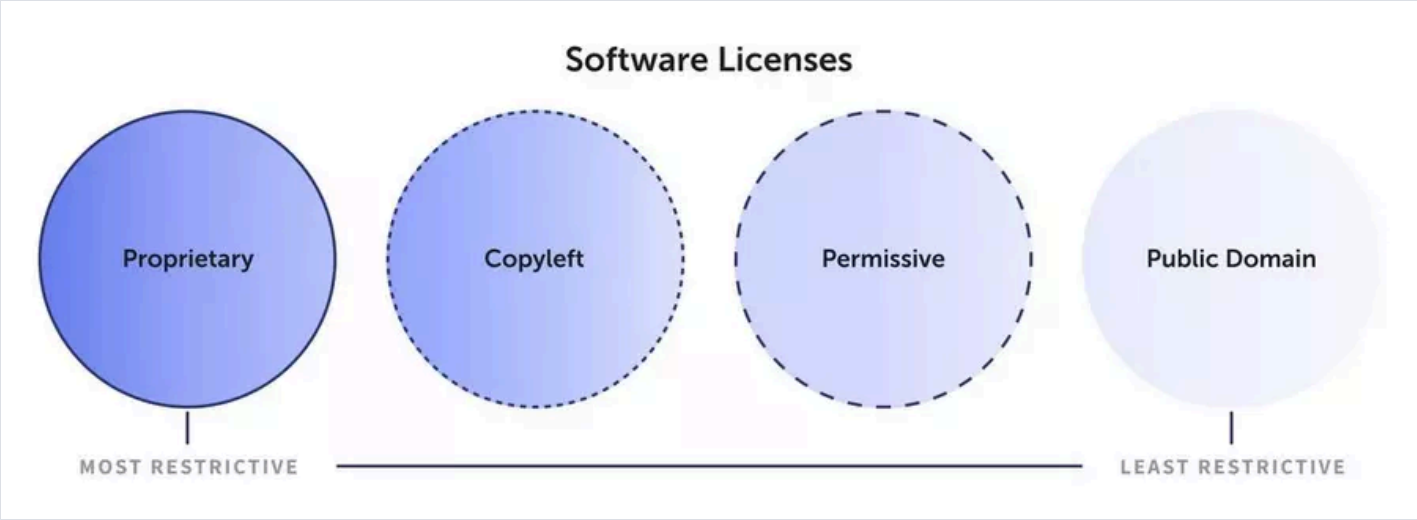
Permissive Licenses

With permissive licenses, users are free to use the software for any purpose, including commercial purposes, and are not required to release the source code or license their modifications under the same terms. This allows for greater flexibility in the use and distribution of the software, but can also lead to the creation of closed, proprietary derivative works.

Permissive licenses are often favored by businesses and individuals who want to use open-source software as a component in their own proprietary software, as they provide greater freedom to use the software in their own projects.



Software licenses have different levels of restrictiveness

Open-Source License Comparison: Main Differences

Open-source licenses are numerous and this is where Open Source Initiative (OSI) plays a crucial role. It is a non-profit organization that promotes and protects open-source software through the development and enforcement of open-source licenses.

These are some of the most popular OSI-approved open-source licenses.

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Copyleft Licenses	Permissive Licenses
GNU General Public License (GPL)	Apache License
GNU LGPLv3.0 (Lesser General Public License)	BSD License
Mozilla Public License	MIT License
Eclipse Public License	Unlicense

Most popular open-source license types

Permissive Open-Source Licenses

- **MIT License**

Created by the Massachusetts Institute of Technology, the MIT license is both one of the most popular licenses and also the most permissive. In contrast to many other open-source licenses, this one is quite short and clear. It states that the source code can be used in any way with the only requirement to preserve copyright and license notices. However, it doesn't expressly grant patent rights which is why some prefer to avoid it. MIT license is used by Babel, .NET, and Rails, to name some of the famous open-source software solutions.

- **BSD License**

Berkeley Software Distribution (BSD) license grants developers similar rights to distribute the work without source code and under different license terms, but asks to include the copyright notice and license text in the copy of the software.

There are several BSD license versions with BSD 3-Clause license being the most popular. It contains a non-endorsement clause that prohibits developers from using the name of the original creators to endorse their modifications.

- **Apache License**

The Apache license was released by Apache Software Foundation (ASF). As a permissive license, it grants developers the flexibility to distribute their work under any preferred license, as long as they credit the initial one and carefully document all modifications made to it. You have the option to keep some of your materials as a closed source and can obtain licenses that allow you to monetize them or even secure a patent. Some famous software solutions protected by Apache license are Kubern
PDF.js, and Swift.

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OPEN SOURCE

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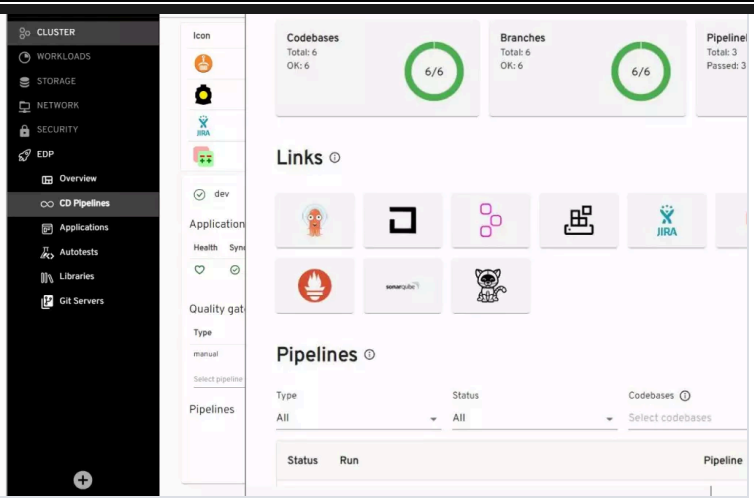

- **The Unlicense**

This is an OSI-approved, public-domain-equivalent license with no conditions. Developers have complete freedom to use the source code any way they want without any obligations. This permissive license is chosen by those who just want to make their software publicly available and don't what to make any rules as to how it can be used but must include a license for legal reasons.

Copyleft Open-Source Licenses

- **GNU General Public License (GPL)**

GNU GPL is one of the first open-source software licenses and is also among the most popular copyleft ones. It belongs to strong copyleft licenses as it requires users to share any modifications they make to the software and to make the source code of their modifications available to the public. Even if the GPL code takes just a small percentage of the new software, the source code of the whole new software has to be available under the same GPL license.

There are several versions of GPL licenses with differences that make them more or less applicable and compatible with other licenses. All the details can be found on the GNU website. For example, GIMP and Bash use this license type.

- **GNU LGPL (Lesser General Public License)**

The GNU Lesser General Public License (LGPL) is a copyleft license that is similar to the GPL but allows users to link the software with another one without requiring them to share the source code of the proprietary software. As such, it falls somewhere between strong and weak copyleft.

- **Mozilla Public License (MPL)**

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As a weak copyleft license, Mozilla Public License claims that if a derivative work modifies code licensed under MPL, it also has to be licensed under the same license. But, if the MPL code is kept in separate files without any modifications made to it, the derivative work can be combined with non-

- Eclipse Public License (EPL)

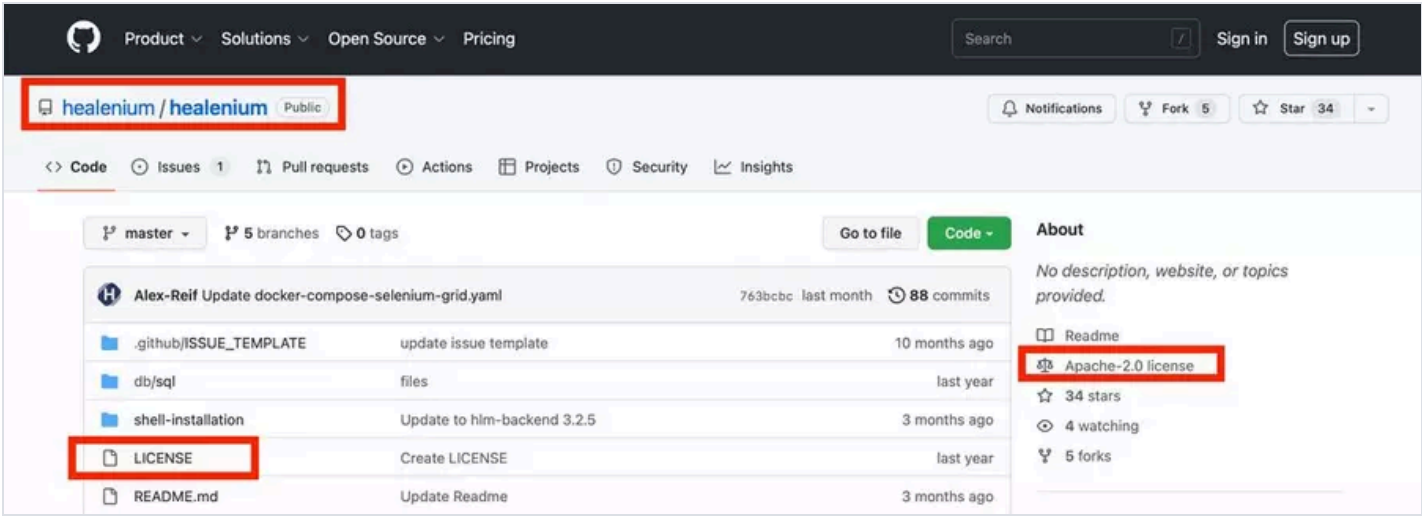
EPL is also a weak copyleft license. If you choose to modify an EPL component and distribute it as part of your program's source code, disclosure of the modified code is necessary. On the other hand, if you provide only object code for distribution, then you must clearly communicate that recipients are able to request the original source code from you along with a description of how to make such requests.

Open-Source License Risks and How to Avoid Them as a User of Open-Source Software

License issues can result in costly litigation that any company would most definitely like to avoid. A lawsuit was recently filed against Microsoft and OpenAI - the company behind the increasingly popular ChatGPT. They are accused of breaching intellectual property laws by utilizing open-source code created by other parties published on GitHub. Microsoft used it to construct and train its Copilot service, which leverages AI for automated software production.

If you use open-source components and libraries to build software, license compliance should be a key concern. Before deciding on open-source software, make sure to check what license is attached to it and if the rest of your code is compatible based on the conditions stated in the license.

The easiest way to do it is by checking it on GitHub. There is information for each project on what license is used and by clicking on it you can get a brief summary of the license as well as the whole text.



License information on GitHub

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How to Choose the Best Open-Source License for Your Code?

So you are ready to share your latest software creation with the world, but unsure which open-source license to opt for? With so many types of open-source licenses out there and almost all written in not

1. If you plan to contribute to an open-source community, the easiest way is to choose the license used by the community. Doing so the project you contribute to the community would fit in perfectly since not all licenses are compatible with one another. Also, some communities restrict the use to only particular licenses. For example, WordPress is released under GPLv2 and that's the only license you can use if you want to contribute to the WordPress community.

2. If you want to make your code as reusable and shareable as possible, to secure the reciprocity you should choose GNU GPLv3. This way others can do almost anything they want with the code apart from distributing closed-source versions.

3. If you want to make it easy and permissive, to let people do with your code whatever they want, including publishing the derived work as proprietary, you should opt for an MIT license.

Conclusion

Open-source licenses play a critical role in the development and distribution of open-source software. They provide a clear framework for the sharing and use of software, allowing for collaboration, innovation, and the creation of high-quality software that benefits the entire community.

There are different types of open-source licenses to choose from, including:

- Strong Copyleft licenses (i.e., GPL)
- Weak Copyleft licenses (i.e., LGPL)
- Permissive licenses (i.e., MIT, BSD)

It's important to choose the right license for your project, but also to understand what open-source licenses to avoid, considering your goals for the software and the potential risks involved. Some risks associated with open-source software include the possibility of others creating closed, proprietary derivative works and the risk of license compatibility issues if the software is combined with other software under different licenses.

In summary, while open-source licenses come with some risks, they also provide numerous benefits and opportunities for collaboration, innovation, and the creation of high-quality software. It's important to carefully consider the right license for your project to ensure that your goals for the software are met.

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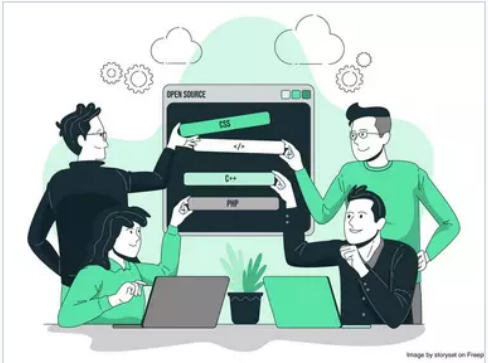

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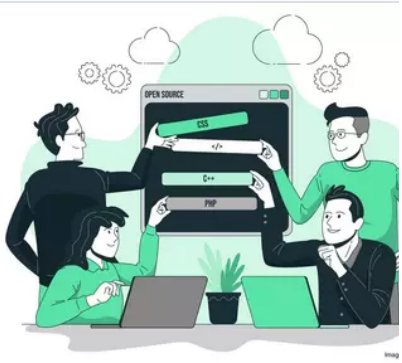

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